

Is success written in the stars? The effect of zodiac sign on the outcome of total knee arthroplasty

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ABSTRACT

AIM: There are 12 signs of the zodiac, each attributed with its own specific personality traits, desires and attitudes. The aim of the study was to evaluate the effect of zodiac sign on patient-reported outcome measures (PROMS) following primary total knee arthroplasty (TKA).

METHOD: Patients undergoing primary TKA during a 2-year period (January 2019 to December 2020) were identified retrospectively. Patient demographics, Oxford Knee Score (OKS), EuroQol 5-dimension (EQ-5D) (baseline, 1 and 2 years) and patient satisfaction scores (1 and 2 years) were collected. Each patient's zodiac sign was assigned from their date of birth.

RESULTS: There were 509 patients (228 males [44.8%] and 281 females [55.2%]) with a mean age of 70.9 years and a mean BMI of 30.3. There were no significant differences in gender ($p=0.712$), age ($p=0.088$), BMI ($p=0.660$), or pre-operative OKS ($p=0.539$). Aries and Gemini (0.366) had the worst and Pisces the best (0.595) pre-operative EQ-5D scores ($p=0.038$). When adjusting for confounding, Aries ($p=0.031$) had a greater improvement in EQ-5D at 1 year, although this was not maintained at 2 years. When adjusting for confounding, Pisceans had significantly less of an improvement in OKS at both 1 ($p=0.022$) and 2 years ($p=0.042$) and also had a significantly lower risk of satisfaction at 2 years (odds ratio 0.41, $p=0.043$).

CONCLUSION: Zodiac sign was associated with outcome following TKA. Pisceans had the best pre-operative EQ-5D scores, but the least improvement in the post-operative joint specific score (OKS) and were less likely to be satisfied, despite achieving an equal improvement in their health-related quality of life (EQ-5D). Aries started with the lowest pre-operative EQ-5D scores but achieved the best scores at 1 year. Our study shows that an individual's zodiac sign may serve as a useful predictive factor for functional outcomes and satisfaction following TKA. However, our findings are the result of multiple testing in a large dataset following a data trawl, and correlation does not necessarily equal causation even in a real-world registry.

KEYWORDS: EQ-5D; Oxford Knee Score; knee; arthroplasty; outcome; zodiac sign; star sign.

Introduction

The position of the sun at the time of a person's birth, placed within one of the 12 sections of the zodiac, is referred to as their zodiac sign.¹ It has been suggested that aspects of an individual's personality and future can be predicated on their zodiac sign,² and from the position of planets, moons and stars. Previous studies evaluating the influence of astrology have noted an association between the severity of myocardial infarction and the presence of the sun,³ as well as a correlation between the position of the moon (lunar phase) and post-surgical pain.⁴

Little explanation can, however, be given for why a particular zodiac sign is associated with certain personality traits, or how these associations were originally derived.^{1,5} Despite the lack of scientific evidence, horoscopes forecasting a person's future based purely on their zodiac sign first appeared in English newspapers in the 1930s

and continue to feature in glossy magazines and on the internet, with birth charts being popular, or essential, in many cultures.^{6,7}

Patient-reported outcome measures (PROMs) are used to assess the efficacy of a surgical intervention from the patient's perspective,⁸ improving the understanding of which outcomes are most relevant for everyday functioning.⁹ The Oxford Knee Score (OKS) is a validated joint-specific PROM¹⁰ that correlates with patient satisfaction and the outcome of total knee arthroplasty (TKA).^{11,12} The EuroQol five dimensions (EQ-5D) is a widely administered PROM for measuring overall health-related quality of life (HRQoL)^{13,14} and is used by many national arthroplasty registries.¹⁵

Despite the optimisation of surgical techniques and implant designs, more than 10% of patients remain dissatisfied following TKA,^{16,17} the outcome of which can also be influenced by factors including age, gender, comorbidities and personality traits.¹⁸⁻²¹ In spite of enduring public fascination, there have,

however, been no previous publications evaluating the association between signs of the zodiac and the outcome of joint replacement surgery. The primary aim of the study was therefore to evaluate the effect of zodiac sign on knee-specific outcome (OKS) following TKA. The secondary aims were to assess the effect of zodiac sign on health-related quality of life (EQ-5D) as well as patient satisfaction up to 2 years following TKA.

Method

Patients undergoing primary TKA during a 2-year period (January 2019 to December 2020) were identified retrospectively from a prospectively compiled arthroplasty database at an Elective Orthopaedic Centre. Only patients undergoing revision TKA were excluded from the database search to keep the sample representative of all patients undergoing primary TKA.

Patient demographics were recorded pre-operatively, and PROMs questionnaires were completed pre-operatively and then at 1 and 2 years. The Oxford Knee Score (OKS) was used to assess the joint-specific outcome measure (primary aim), the EuroQol 5-dimension (EQ-5D) to assess health-related quality of life, and patient satisfaction was assessed. Each patient's zodiac sign was assigned from their date of birth, taking into account leap years.

The OKS consists of 12 questions, assessed using a Likert scale, with values from 0 to 4. A summative score is calculated with 48 being the best possible score (least symptomatic) and 0 the worst possible score (most symptomatic).^{22,23} The minimal clinically important difference (MCID) is the smallest change of score, deemed to be of clinical importance, and has been defined as 5-points following TKA.²⁴

The EQ-5D general health questionnaire evaluates five dimensions including mobility, self-care, usual activities, pain/discomfort and anxiety/depression.¹³ The 3L version of the EQ-5D questionnaire was used in the current study, with the responses to the five domains being recorded at three levels of severity. The health state can then be reported based on a three-digit code for each domain, resulting in 243 possible health states.²⁵ Each health state was converted to a single summary index by applying a weighting, specific to the United Kingdom (UK) population, based on a time trade-off technique. The index is on a scale of -0.584 to 1, with 1 being perfect health, 0 being dead and negative values representing a state perceived as being worse

than death. A MCID has not yet been defined following TKA, but a review found the MCID to vary from 0.03 to 0.54 depending on the orthopaedic intervention being assessed.²⁶ For the current study 0.1 was defined as the MCID.

Patient satisfaction was assessed by asking the patient to rate their “*overall satisfaction with the outcome of your operation?*” using a visual analogue scale (VAS)¹³ from 0 (not satisfied) to 100 (very satisfied). Patients scoring 50 or more were defined as being satisfied, with those scoring under 50 being dissatisfied.

Statistical analysis

Statistical analysis was performed using Statistical Package for Social Sciences version 17.0 (SPSS Inc., Chicago, IL, USA). Simple descriptive analysis was undertaken. Student's t-Tests, paired and unpaired, or analysis of variance (ANOVA) were used to compare linear variables between groups with Bonferroni correction. *Post hoc* testing was undertaken between star signs using a Bonferroni correction. Dichotomous variables were assessed using a Chi-squared test. Linear regression analyses were used to assess for variables independently associated with improvement in the OKS and EQ-5D utility at 1 and 2 years, when adjusting for confounding using step-wise methodology. Logistic regression analyses were used to assess for variables independently associated with patient satisfaction at 1 and 2 years, when adjusting for confounding using forward conditional methodology. Each star sign was included in the models as a categorical variable, with no base star sign identified on unadjusted analysis; the “rest” of the cohort was used as the reference group e.g., Taurus versus all other star signs. A p-value of <0.05 was defined as statistically significant.

A power calculation was performed using the OKS as the primary outcome measure, with change in the score relative to baseline at 1-year as the end point. MCID of 5²⁴ was used for the OKS with a standard deviation of 10, giving an effect size of 0.5, an alpha of 0.05 and a power of 95%—for 12 groups (the zodiac signs) using a one-way ANOVA, 120 patients would be required.

No additional patient contact was required and the project was performed as a service evaluation, without the need for formal ethical approval. The project was registered with the institution's Audit Department and was conducted in accordance with the Declaration of Helsinki and the guidelines for Good Clinical Practice.

Results

Cohort

There were 228 (44.8%) males and 281 (55.2%) females with a mean age of 70.9 years (range 44 to 94). The mean pre-operative OKS was 21.8 (standard deviation [SD] 8.1) and the mean pre-operative EQ-5D was 0.474 (SD 0.036). There were no significant differences according to Zodiac sign for age, sex, BMI or OKS pre-operatively (Table 1). Zodiac sign was, however, related to pre-operative HRQoL, with Aries and Gemini having the worst (lowest) EQ-5D utility scores and Pisces having the best (highest) (Table 1). On *post hoc* testing the only significant difference was between Aquarius and Taurus (difference 0.229, 95% CI 0.000 to 0.458, $p=0.049$).

Primary aim: knee-specific outcome following TKA

There was a significant ($p<0.001$, paired t-Tests) post-operative improvement, relative to the pre-operative OKS for all Zodiac signs at both 1 and 2 years (changes in Table 2). On unadjusted analysis there was no significant difference according to Zodiac sign at 1 ($p=0.196$ ANOVA) or 2 years ($p=0.392$ ANOVA) (Table 2). When adjusting for confounding factors (age, gender, BMI), Pisces was independently associated with a significantly worse improvement at both 1 (difference -2.8 points, $p=0.022$) and 2 years (-2.5 points, $p=0.035$) (Table 3). These were, however, less than the MCID (5 points).

Secondary aim: HRQoL

There was a significant ($p<0.001$, paired T-tests) improvement in the post-operative EQ-5D at both 1 and 2 years compared to baseline for all patients following TKA (changes in Table 4). On un-adjusted analysis there was no significant difference in HRQoL according to zodiac sign at 1 year ($p=0.113$ ANOVA) but by 2 years Taureans had the least improvement in EQ-5D, and Aquarians the greatest ($p=0.023$ ANOVA) (Table 4). However, when adjusting for confounding factors (age, gender, BMI) Aries was the only sign associated with a greater improvement at 1 year, although this was lower than the MCID and was not maintained at 2 years (Table 3).

Secondary aim: patient satisfaction

Overall, patient satisfaction was 93.9% at 1 year and 91.9% at 2 years. On unadjusted analysis there was no association between zodiac sign and

satisfaction at either 1 ($p=0.272$) or 2 years ($p=0.409$) (Table 5), but, when adjusting for confounding factors (age, gender, BMI), Pisces was associated with a significantly lower risk of satisfaction at 2 years (odds ratio 0.41, $p=0.043$) (Table 6).

Discussion

Zodiac sign was associated with post-operative knee-specific function and patient satisfaction following TKA. Pisceans started with the highest pre-operative EQ-5D scores, but experienced significantly worse improvement in their post-operative joint-specific score (OKS) at both 1 (difference -2.8 points, $p=0.022$) and 2 years (-2.5 points, $p=0.035$). Whether this is clinically significant is not clear since the differences were lower than the MCID of 5 points, although the authors of the Oxford Knee Score suggest that the MCID may actually be lower, at 3 points. Pisceans were also less likely to be satisfied with the outcome of their knee, which supports the potential clinically significant effect they experience due to the lower improvement in their knee-specific outcome.

Pisces is the final sign of the zodiac (19 February to 20 March), spanning 330 to 360 degrees of celestial longitude,²⁷ and along with Scorpio and Cancer is a water sign. Its symbol is two fish tied to one another, but swimming in opposite directions amid shifting emotional desires, plagued by conflict and extremes of temperament.⁷ The great advocate of science and astrology, Cosmopolitan,⁷ concluded that Pisceans feel everything more deeply and are thin-skinned in a cruel world.

Abdel-Khalek and Lester²⁸ found Pisceans to be significantly more anxious than other star signs, a psychological trait associated with poorer OKS at 1 year following TKA.²⁹ Pisceans also tend to be more sensitive and emotional than other star signs,³⁰ consistent with other studies reporting that individuals with greater emotional instability were significantly more likely to be dissatisfied, to experience greater psychological distress and to have poorer coping skills in times of stress.^{21,31}

In contrast to the poor OKS and satisfaction scores, Pisceans enjoyed an equal improvement in their HRQoL, which begs the question—are these findings real, or is this simply a red herring?

In the current study, three different outcome measures were assessed at three different time points, six different statistical tests were performed for the 12 zodiac signs and three regression models

were undertaken. Pisces was caught in the net at several time points in different statistical tests and we therefore believe that the results represent a real association. Whether there is causation is a different matter, and there is the possibility that our findings may be due to multiplicity or chance given the number of analyses that were performed, so our findings on Pisces must therefore be taken with a pinch of salt (and possibly a dash of vinegar!).

In contrast to Pisces, Aries is the first sign of the zodiac (21 March to 19 April), spanning the first 30 degrees of celestial longitude,²⁷ and along with Sagittarius and Leo is a fire sign. The symbol is the ram, with a brave, competitive, tenacious ability to climb to the top of the mountain.⁷ Perhaps these attributes help Aries to overcome the initial vagaries of TKA, achieving the greatest post-operative improvement in HRQoL at 1 year. Unsurprisingly, due to the effort involved, they might be expected to run out of steam, as this improvement was not maintained at 2 years.

Limitations of the current study include its retrospective nature, but despite this there were no significant pre-operative differences in patient demographics (age, sex, BMI) or OKS according to zodiac sign, all of which are known to be independent predictors of pain and functional outcome following TKA.³² Another limitation is the lack of other comorbidity data such as mental health status, diabetes, severity of osteoarthritis and knee malalignment, which are also factors associated with poorer outcomes following TKA.^{33–38} Additionally, the study was conducted during the pre-COVID and COVID pandemic period, however, any impact on outcomes was expected to be evenly and randomly distributed across all patient groups.

Our study evaluated the influence of the position of the sun at the time of a person's birth, since this is how astrological dates are determined in the United Kingdom (UK). However, it should be noted that not all astrological calendars use the sun. For example, the Māori calendar (Maramataka) uses different dates by following the movements of the

moon over a typical month and year, and there is evidence that moon phases may affect outcomes following surgery.⁴ Furthermore, birthplace and religion may have an impact on an individual's astrological sign, and this was not taken into account in the current study. This study may therefore need to be adapted to the local calendar, and further research is required into the effect of different astrological systems on surgical outcomes, accounting for other variables such as birthplace and religion.

The validity of zodiac sign astrology must also be considered. It has been described as a pseudoscience, with little evidence for the existence of relevant relationships between zodiac sign and individual personality traits.³⁹ The occasional associations in personality between zodiac signs observed in studies^{30,40} are thought likely to be accounted for by self-attributions, with individuals believing in astrology attributing personality traits consistent with their zodiac sign to themselves.¹

Nevertheless, there is no point carping about it, or perching on the fence, and having had the opportunity to mullet over the current study did reveal some significant findings, which could be of importance to both physicians and surgeons.

Conclusion

Zodiac sign was associated with outcome following TKA. Pisceans had less improvement in the joint-specific score (OKS) that persisted at 2 years and were less likely to be satisfied with their TKA, despite which they enjoyed an equal improvement in their quality of life. Aries showed the greatest improvement in EQ-5D 1 year after TKA, but this benefit was not maintained at 2 years. Our study shows that an individual's zodiac sign may serve as a useful predictive factor for functional outcomes and satisfaction following TKA. However, our findings are the result of multiple testing in a large dataset following a data trawl, and correlation does not necessarily equal causation even in a real-world registry.

Table 1: Pre-operative patient demographics, Oxford Knee Score (OKS) and EuroQol 5-dimension (EQ-5D), according to zodiac sign.

Star sign	Gender		P-value*	Age		P-value*	BMI		P-value*	OKS		P-value*	EQ-5D		P-value*
	Male	Female		Mean	SD		Mean	SD		Mean	SD		Mean	SD	
Aquarius	17 (45.9%)	20 (54.1%)	0.712	72.8	7.7	0.088	29.9	7.5	0.660	24.3	8.6	0.539	0.557	0.281	0.038
Aries	19 (48.7%)	20 (51.3%)		72.6	7.6		30.8	6.3		20.8	7.7		0.366	0.310	
Cancer	13 (36.1%)	23 (63.9%)		69.7	9.1		29.5	4.4		20.3	7.4		0.459	0.291	
Capricorn	27 (56.3%)	21 (43.8%)		72.6	7.9		31.2	5.2		21.7	9.3		0.471	0.330	
Gemini	16 (34.8%)	30 (65.2%)		70.7	9.3		29.4	4.6		20.0	7.2		0.366	0.314	
Leo	24 (42.9%)	32 (57.1%)		71.7	7.0		31.7	6.4		22.8	7.4		0.474	0.283	
Libra	20 (47.6%)	22 (52.4%)		71.8	6.3		30.2	4.5		21.7	7.7		0.483	0.292	
Pisces	23 (46%)	27 (54%)		70.7	8.0		30.6	5.0		22.4	6.8		0.595	0.234	
Sagittarius	14 (36.8%)	24 (63.2%)		68.9	6.1		30.5	6.0		22.6	8.9		0.492	0.334	
Scorpio	17 (41.5%)	24 (58.5%)		70.9	8.2		30.1	5.5		22.3	7.8		0.500	0.278	
Taurus	22 (50%)	22 (50%)		68.0	7.6		30.1	4.9		20.7	9.3		0.433	0.351	
Virgo	16 (50%)	16 (50%)		71.3	7.8		29.5	3.9		21.4	8.4		0.441	0.332	
Total	228 (44.8%)	281 (55.2%)		70.9	7.8		30.3	5.5		21.8	8.1		0.474	0.306	

* ANOVA corrected for multiple testing for comparison of change between all zodiac signs.

Table 2: Change in Oxford Knee Score (OKS) at 1-year and 2-years post-operative compared to pre-operative values, according to zodiac sign.

Star sign	N	One year				Two years			
		Mean	95% CI	P-value*	P-value**	Mean	95% CI	P-value*	P-value**
Aquarius	37	15.4	12.2 to 18.7	<0.001	0.196	18.2	15.0 to 21.3	<0.001	0.392
Aries	39	18.9	15.8 to 22.1	<0.001		19.8	17.4 to 22.2	<0.001	
Cancer	36	17.8	14.8 to 20.7	<0.001		19.2	15.8 to 22.6	<0.001	
Capricorn	48	18.1	15.6 to 20.5	<0.001		17.0	14.5 to 19.6	<0.001	
Gemini	46	15.5	12.5 to 18.4	<0.001		17.3	14.6 to 20.1	<0.001	
Leo	56	15.3	12.7 to 17.8	<0.001		16.9	14.6 to 19.2	<0.001	
Libra	42	15.9	13.0 to 18.7	<0.001		16.5	14.2 to 18.9	<0.001	
Pisces	50	13.7	11.0 to 16.4	<0.001		14.6	12.0 to 17.2	<0.001	
Sagittarius	38	16.8	13.7 to 19.8	<0.001		16.1	13.0 to 19.1	<0.001	
Scorpio	41	14.8	11.8 to 17.8	<0.001		15.5	12.6 to 18.4	<0.001	
Taurus	44	17.8	15.3 to 20.3	<0.001		17.2	14.1 to 20.3	<0.001	
Virgo	32	18.1	14.8 to 21.5	<0.001		16.4	12.6 to 20.2	<0.001	
Total	509	16.4	15.6 to 17.2	<0.001		17.0	16.2 to 17.8	<0.001	

* Paired Student's t-Test for change in score relative to baseline.

**ANOVA corrected for multiple testing for comparison of change between all zodiac signs.

Table 3: Linear regression models for factors associated with change in Oxford Knee Score (OKS) or EuroQol 5-dimension (EQ-5D) at 1-year and 2-years post-operative compared to pre-operative values.

PROM & timepoint	Variables	B	Lower	Upper	P-value
OKS 1 year	OKS baseline	-0.6	-0.7	-0.5	0.000
	BMI	-0.2	-0.3	0.0	0.015
	Pisces	-2.8	-5.2	-0.4	0.022
OKS 2 years	OKS baseline	-0.6	-0.7	-0.5	0.000
	BMI	-0.3	-0.4	-0.1	0.000
	Pisces	-2.5	-4.8	-0.2	0.035
	Male1Female2	-1.4	-2.8	0.0	0.042
EQ-5D 1 year	EQ-5D baseline	-0.869	-0.952	-0.786	0.000
	OKS baseline	0.006	0.003	0.009	0.000
	Age	0.003	0.001	0.006	0.006
	Aries	0.076	0.007	0.145	0.031
EQ-5D 2 years	EQ-5D baseline	-0.864	-0.951	-0.776	0.000
	OKS baseline	0.005	0.001	0.008	0.005
	BMI	-0.005	-0.009	-0.001	0.008

Table 4: Change in EuroQol 5-dimension (EQ-5D) utility index at 1-year and 2-years post-operative compared to pre-operative values, according to zodiac sign.

Star sign	N	1 year				2 years			
		Mean	95% CI	P-value*	P-value**	Mean	95% CI	P-value*	P-value**
Aquarius	37	0.337	0.213 to 0.461	<0.001	0.113	0.460	0.338 to 0.582	<0.001	0.023
Aries	39	0.411	0.304 to 0.519	<0.001		0.410	0.299 to 0.521	<0.001	
Cancer	36	0.333	0.219 to 0.448	<0.001		0.382	0.260 to 0.503	<0.001	
Capricorn	48	0.322	0.236 to 0.407	<0.001		0.320	0.237 to 0.402	<0.001	
Gemini	46	0.292	0.201 to 0.383	<0.001		0.316	0.207 to 0.426	<0.001	
Leo	56	0.327	0.245 to 0.409	<0.001		0.372	0.292 to 0.451	<0.001	
Libra	42	0.279	0.182 to 0.376	<0.001		0.288	0.195 to 0.382	<0.001	
Pisces	50	0.281	0.191 to 0.370	<0.001		0.257	0.161 to 0.353	<0.001	
Sagittarius	38	0.343	0.245 to 0.441	<0.001		0.337	0.247 to 0.427	<0.001	
Scorpio	41	0.230	0.119 to 0.344	<0.001		0.261	0.167 to 0.355	<0.001	
Taurus	44	0.222	0.163 to 0.281	<0.001		0.224	0.150 to 0.298	<0.001	
Virgo	32	0.430	0.309 to 0.552	<0.001		0.406	0.276 to 0.536	<0.001	
Total	509	0.313	0.286 to 0.341	<0.001		0.332	0.303 to 0.360	<0.001	

*Paired Student's t-Test for change in score relative to baseline.

**ANOVA corrected for multiple testing for comparison of change between all zodiac signs.

Table 5: Patient satisfaction with their total knee arthroplasty (TKA) at 1-year and 2-years post-operative, according to zodiac sign.

Star sign	1 year		P-value	2 years		P-value
	Satisfied	Dissatisfied		Satisfied	Dissatisfied	
Aquarius	35 (95%)	2 (5%)	0.272	34 (92%)	3 (8%)	0.409
Aries	38 (97%)	1 (3%)		37 (95%)	2 (5%)	
Cancer	35 (97%)	1 (3%)		33 (92%)	3 (8%)	
Capricorn	46 (96%)	2 (4%)		44 (92%)	4 (8%)	
Gemini	40 (87%)	6 (13%)		41 (89%)	5 (11%)	
Leo	54 (96%)	2 (4%)		56 (100%)	0 (0%)	
Libra	40 (95%)	2 (5%)		40 (95%)	2 (5%)	
Pisces	45 (90%)	5 (10%)		42 (84%)	8 (16%)	
Sagittarius	35 (92%)	3 (8%)		34 (90%)	4 (11%)	
Scorpio	38 (93%)	3 (7%)		37 (90%)	4 (10%)	
Taurus	44 (100%)	0 (0%)		40 (91%)	4 (9%)	
Virgo	28 (88%)	4 (13%)		30 (94%)	2 (6%)	
Total	478 (94%)	31 (6%)		468 (92%)	41 (8%)	

Table 6: Logistic regression models for factors associated with patient satisfaction at 6 months, 1-year and 2-years post-operative.

		OR	Lower	Upper	P-value
1 year	EQ-5Dbaseline	7.00	2.28	21.52	0.001
2 years	Age	1.05	1.01	1.09	0.015
	OKS baseline	1.08	1.03	1.13	0.001
	Pisces	0.41	0.17	0.97	0.043

COMPETING INTERESTS

The authors declare that they have no competing interests.

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All available data are provided. Additional data, if needed, may be made available from the corresponding author on reasonable request.

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